8C 1 TRENDS IN GROUP 7

SPECIFICATION REFERENCE
8.24 8.28

LEARNING OBJECTIVES

- Understand reasons for the trends in melting and boiling temperatures, physical state at room temperature, electronegativity and reactivity down the group.
- Be able to make predictions about fluorine and astatine and their compounds, in terms of knowledge of trends in halogen chemistry.

INTRODUCTION TO THE GROUP 7 ELEMENTS

Group 7 of the Periodic Table contains five elements. These elements are often known as halogens and they all form salts called halides. The term 'halogen' comes from Greek and means 'salt producer'.

When considering the properties of the group, the elements at the top and bottom of the group (fluorine and astatine) are often ignored. Fluorine is ignored because it sometimes behaves differently from chlorine, bromine and iodine, and astatine is ignored because (like radium in Group 2) it only exists as radioactive isotopes.

Table A shows some information about the Group 7 elements.

ELEMENT	STATE AT ROOM TEMPERATURE	MELTING TEMPERATURE / °C	BOILING TEMPERATURE / °C	ELECTRONEGATIVITY
Fluorine	gas	−220	−188	4.0
Chlorine	gas	−101	−35	3.0
Bromine	liquid	−7	59	2.8
Iodine	solid	114	184	2.5
Astatine	solid	302	337	2.2

table A Physical properties of Group 7 elements.

TRENDS IN MELTING AND BOILING TEMPERATURE

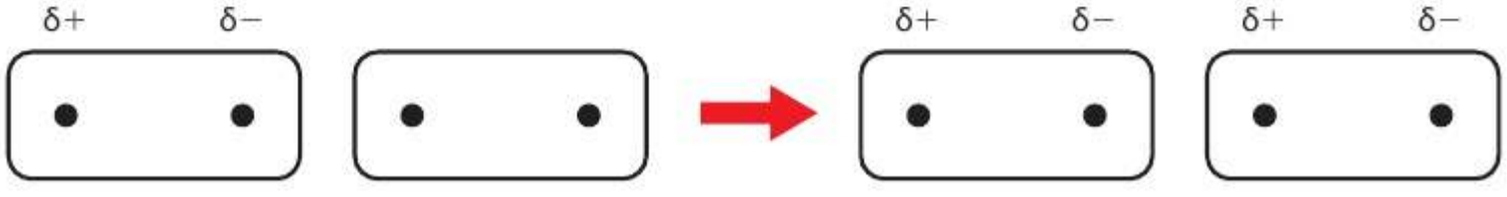
All of the halogens exist as diatomic molecules, so their melting and boiling temperatures depend on the strengths of the intermolecular forces of attraction between these molecules. We looked at these forces, known as London forces, in **Topic 7**. Here is a reminder of how they happen.

As the two atoms in the diatomic molecule are identical, the pair of electrons forming the covalent bond between them is shared equally between the two atoms. This means that the halogen molecules are non-polar, at least on average. However, as the positive charges of the protons in the two nuclei are in fixed positions, but the electron density in a halogen molecule continuously fluctuates, sometimes the centres of positive and negative charge do not coincide. This situation results in a temporary dipole, which is often referred to as an instantaneous dipole.

We will use a rectangular shape to represent a halogen molecule. The two dots represent the nuclei of the two halogen atoms. When two molecules are close together, you would expect no interaction between the two because they are both non-polar.



If the molecule on the left becomes an instantaneous dipole, then it will cause an induced dipole in the molecule on the right. This results in a force of attraction between the two molecules:



This force of attraction is described as an instantaneous dipole–induced dipole attraction, and these are the intermolecular forces of attraction that exist between halogen molecules. These weak forces increase as the number of electrons and therefore the size of the electron cloud increases. The forces increase in strength down Group 7 as the number of electrons in the molecules increases. This explains the increase in both melting temperature and boiling temperature down Group 7.

EQUATIONS FOR CHANGES OF STATE

You are probably used to writing equations for chemical changes, but we use equations for physical changes less often. As well as using the correct state symbols, it is important to remember to write the formulae (not the symbols) of the halogens.

When bromine is left at room temperature, it gives off brown vapour, as its boiling point (59 °C) is not much higher than room temperature. The equation for this change is:

$$Br_2(l) \rightarrow Br_2(g)$$

When iodine is warmed, most of it changes directly into a vapour without melting. This change is called **sublimation**. The equation for this change is:

$$I_2(s) \rightarrow I_2(g)$$

TREND IN ELECTRONEGATIVITY

We have already seen in **Topic 3B** that electronegativity is the ability of an atom to attract the pair of electrons in a covalent bond. The 0–4 scale devised by Linus Pauling is still used. The electronegativity of an atom depends on:

- its nuclear charge – the bigger the nuclear charge, the higher the electronegativity
- the distance between the nucleus and the bonding pair of electrons – the shorter the distance, the higher the electronegativity
- the shielding effect of electrons in inner energy levels – the fewer energy levels, the higher the electronegativity.

The electronegativity of the Group 7 elements is the highest of any group in the Periodic Table. The electronegativity of fluorine is the highest of all elements.

TREND IN REACTIVITY

Fluorine is an extremely reactive element, and reactivity decreases down Group 7. Because of their high electronegativity, most reactions of the halogens involve them acting as oxidising agents and gaining electrons to form negative ions or becoming the slightly negative ($\delta-$) part of a polar molecule. The decreasing reactivity down the group can therefore be explained by reference to the same factors used to explain the decreasing electronegativity down the group.

We will see examples of reactions in **Topics 8C.2 to 8C.4**.

LEARNING TIP

Write a brief summary to help you understand the difference between intermolecular forces involving permanent dipoles and temporary dipoles.

CHECKPOINT

1. Why does bromine have a higher boiling temperature than chlorine?

2. Why is fluorine the most electronegative element of all?

SUBJECT VOCABULARY

sublimation the process of a solid changing directly into a vapour without melting

8C 2 REDOX REACTIONS IN GROUP 7

SPECIFICATION REFERENCE
8.25 8.26

LEARNING OBJECTIVES

- Understand the trend in reactivity of Group 7 elements in terms of the redox reactions of chlorine, bromine and iodine with halide ions in aqueous solution, followed by the addition of an organic solvent.
- Understand, in terms of changes in oxidation number, oxidation reactions of the halogens with Group 1 and 2 metals and disproportionation reactions of chlorine with water, and with cold and hot alkali.

REACTIONS WITH METALS IN GROUPS 1 AND 2

There are 12 elements in Groups 1 and 2, and 5 elements in Group 7, so there are 60 possible reactions to consider. You do not need to know details of all of these reactions, but here are some useful generalisations that you should know.

- Reactions are most vigorous between elements at the bottom of Groups 1 and 2, and elements at the top of Group 7. The most vigorous reaction should be between caesium (or francium) and fluorine, and the least vigorous between beryllium and iodine (or astatine).
- The products of these reactions are salts, ionic solids that are usually white.
- All of these reactions involve electron transfer to the halogen, so they are redox reactions in which the halogen acts as an oxidising agent.
- The oxidation number of the halogen decreases from 0 to −1, and the oxidation number of the metal increases from 0 to +1 or +2, depending on the group.

Here are a couple of sample equations:

- lithium reacting with chlorine $2Li + Cl_2 \rightarrow 2LiCl$
- barium reacting with bromine $Ba + Br_2 \rightarrow BaBr_2$

HALOGEN/HALIDE DISPLACEMENT REACTIONS

A more reactive halogen can displace a less reactive halogen from one of its compounds. So:

- chlorine displaces bromine and iodine
- bromine displaces iodine but not chlorine
- iodine does not displace either chlorine or bromine.

These reactions occur in aqueous solution, so any reaction that occurs is indicated by a colour change. One problem in interpreting colour changes in these reactions is the similarity of some colours and the variation in colour with concentration. For example, bromine in its liquid state is red-brown, but bromine dissolved in water might be orange or yellow, depending on the concentration. Iodine dissolved in water may also appear brown at some concentrations.

When doing these reactions, it is a good idea to add an organic solvent (such as cyclohexane) after the reaction, and then shake the tube. Halogens are more soluble in cyclohexane than in water, so the halogen dissolves in the organic upper layer, where its colour can more easily be seen. **Fig A** shows the colours of the halogens dissolved in cyclohexane.



▲ **fig A** Look at the upper layer in each tube. The pale green colour of chlorine does not change much, the orange colour of bromine looks a bit darker, but the colour of iodine changes to purple or violet in cyclohexane.

Here are some sample equations:

- chlorine displacing bromine $Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$
- bromine displacing iodine $Br_2 + 2I^- \rightarrow 2Br^- + I_2$

These equations are examples of redox reactions. The reacting halogen decreases its oxidation number from 0 to −1, and the reacting halide increases its oxidation number from −1 to 0.

The decreasing reactivity of chlorine, bromine and iodine in the reactions above can be explained using the same factors as in **Topic 8C.1**. Chlorine is the most reactive of the three because:

- it is the smallest atom, so the incoming electron gets closer to, and is more attracted by, the protons in the nucleus
- it has the smallest number of complete inner energy levels of electrons, so the incoming electron experiences the least repulsion.

DISPROPORTIONATION REACTIONS OF CHLORINE

Disproportionation is a more unusual type of reaction. In this reaction, one element undergoes both oxidation and reduction at the same time. We will look at three examples of this type of reaction, all involving chlorine.

CHLORINE WITH WATER

When chlorine is added to water, it dissolves to form a solution that is sometimes called 'chlorine water' (just as 'bromine water' is used to refer to bromine dissolved in water). Some of the dissolved chlorine also reacts to form a mixture of two acids.

You are familiar with one of the acids, hydrochloric acid, but the other acid is chloric(I) acid (its old name is hypochlorous acid). Its formula is shown in the equation below as HClO, but HOCl is also commonly used. Both acids are colourless solutions, so there is no visible change during the reaction.

The disproportionation that occurs can be shown using oxidation numbers:

$Cl_2 + H_2O \rightarrow HCl + HClO$

0 −1 chlorine is reduced

0 +1 chlorine is oxidised

The addition of chlorine to disinfect water for drinking purposes has saved countless lives, and continues to do so today. It kills the pathogens responsible for water-borne diseases such as cholera.



▲ **fig B** Chlorine can reduce the risk of transmitting infections in public swimming pools.

CHLORINE WITH COLD ALKALI

When chlorine is added to cold dilute aqueous sodium hydroxide, it reacts to form the salts of the acids in the equation above. These salts are sodium chloride and sodium chlorate(I), which is also known as sodium hypochlorite.

Again, the disproportionation that occurs can be shown using oxidation numbers:

$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$

0 −1 chlorine is reduced

0 +1 chlorine is oxidised

The sodium chlorate(I) formed is also a disinfectant, but it is mainly known for its bleaching action. It is used extensively in industry and is the active ingredient in household bleach.

CHLORINE WITH HOT ALKALI

When chlorine is added to hot concentrated sodium hydroxide solution, it reacts to form sodium chloride and a different product, sodium chlorate(V).

Again, the disproportionation that occurs can be shown using oxidation numbers:

$$3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$$

0 −1 chlorine is reduced

0 +5 chlorine is oxidised

The sodium chlorate(V) formed is also used in bleaching, and as a weed killer.

Bromine and iodine react in similar ways.

REACTIONS OF FLUORINE AND ASTATINE

If you are asked to predict reactions of fluorine and astatine that you are not familiar with, then you can use information in this topic to help you. For example, you could write an equation to represent the reaction between sodium and astatine, based on your knowledge of the reaction between sodium and iodine.

LEARNING TIP

Read the explanation for chlorine being more reactive than the other halogens (except for fluorine) at the end of **Topic 8C.1**. Now try to explain why astatine is the least reactive halogen.

CHECKPOINT

1. (a) Write a chemical equation for the reaction between chlorine and potassium iodide.

(b) Write an ionic equation for the reaction between bromine and sodium astatide.

2. Write equations and name the products of the reactions between bromine and:

(a) cold dilute aqueous sodium hydroxide

(b) hot concentrated sodium hydroxide solution.

SUBJECT VOCABULARY

disproportionation reaction a reaction involving the simultaneous oxidation and reduction of an element in a single species

8C 3 REACTIONS OF HALIDES WITH SULFURIC ACID

SPECIFICATION REFERENCE 8.27(i)

LEARNING OBJECTIVES

- Understand the reactions of solid Group 1 halides with concentrated sulfuric acid, to illustrate the trend in reducing ability of the hydrogen halides.

REDOX REACTIONS AGAIN

In **Topics 8A.3**, **8C.1** and **8C.2**, you have learnt about several examples of the halogens acting as oxidising agents, and you know that this oxidising power decreases down the group.

In this topic, we will look at reactions of halide *ions*, not halogen *molecules*. It is important to realise that in these reactions, the halides act as reducing agents, and that the trend is different. This is shown in **table A**.

OXIDISING POWER	HALOGEN		HALIDE		REDUCING POWER
High ↑	fluorine	F_2	fluoride	F^-	Low ↓
	chlorine	Cl_2	chloride	Cl^-	
	bromine	Br_2	bromide	Br^-	
	iodine	I_2	iodide	I^-	
Low	astatine	At_2	astatide	At^-	High

table A Notice that the decreasing trend down the group in oxidising power of halogens goes with the increasing trend in reducing power of halides.

The reducing action of halide ions can be represented by this general half-equation:

$$2X^- \rightarrow X_2 + 2e^-$$

Sulfuric acid is of course an acid, but when concentrated it contains very few ions. We can write an equation for its partial ionisation:

$$H_2SO_4 \rightleftharpoons H^+ + HSO_4^-$$

Note the reversible arrow. The position of this equilibrium lies well to the left in the concentrated acid, and this first ionisation is far from complete. The second ionisation, to produce sulfate ions, occurs only to a small extent in the concentrated acid:

$$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$$

Sulfuric acid, especially when concentrated, can act as an oxidising agent as well as an acid. When it acts as an oxidising agent, it is reduced, but the extent of its reduction and the products formed depend on the species being oxidised.

The three possible reduction products are:

- sulfur dioxide
- sulfur
- hydrogen sulfide.

Three different half-equations can be written to represent its oxidising action. Note the change in oxidation number of the sulfur in each case:

Half-equation 1. $H_2SO_4 + 2H^+ + 2e^- \rightarrow 2H_2O + SO_2$
$+6 \qquad\qquad +4$

Half-equation 2. $H_2SO_4 + 6H^+ + 6e^- \rightarrow 4H_2O + S$
$+6 \qquad\qquad 0$

Half-equation 3. $H_2SO_4 + 8H^+ + 8e^- \rightarrow 4H_2O + H_2S$
$+6 \qquad\qquad -2$

These half-equations may look complicated at first, but you should be able to see the pattern.

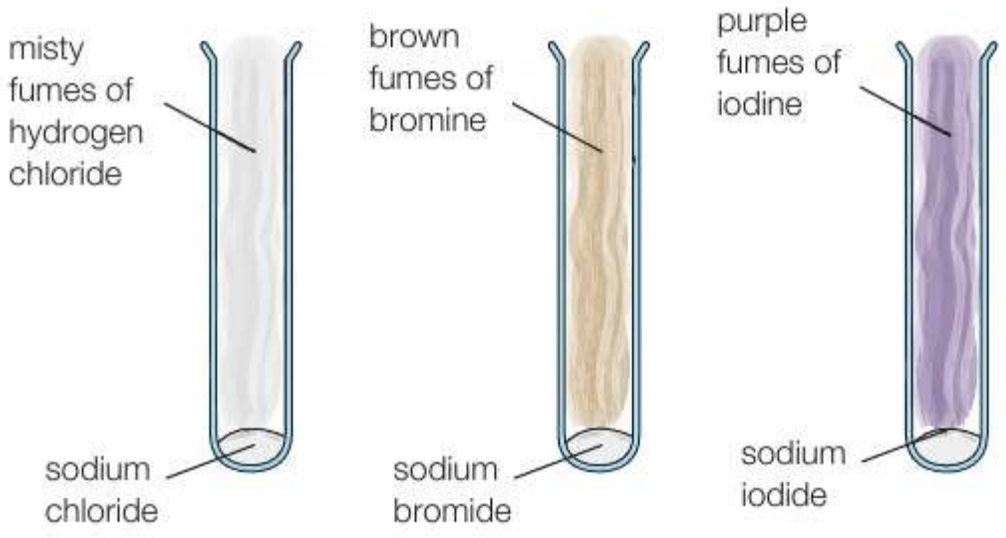
- In half-equation 1, the decrease in oxidation number (+6 to +4) is 2, which is the same as the numbers of H^+ ions and electrons in the equation.
- The pattern is similar in the other two equations. The decrease in oxidation number of the sulfur is the same as the numbers of H^+ ions and electrons in the half-equation (6 in reaction 2, 8 in reaction 3).

OBSERVATIONS AND PRODUCTS

Table B shows typical observations made, and products formed, when concentrated sulfuric acid is added to three sodium halides.

HALIDE	OBSERVATIONS	PRODUCTS	
NaCl	misty fumes	hydrogen chloride	HCl
NaBr	misty fumes brown fumes colourless gas with choking smell	hydrogen bromide bromine sulfur dioxide	HBr Br_2 SO_2
NaI	misty fumes purple fumes or black solid colourless gas with choking smell yellow solid colourless gas with rotten egg smell	hydrogen iodide iodine sulfur dioxide sulfur hydrogen sulfide	HI I_2 SO_2 S H_2S

table B Observations and products formed when concentrated sulfuric acid is added to three sodium halides.



▲ **fig A** From left to right: tubes containing hydrogen chloride (formed from sodium chloride), bromine (formed from sodium bromide) and iodine (formed from sodium iodide).

The tube on the left in **fig A** shows no colour. These are the misty fumes of hydrogen chloride, formed from sodium chloride. The tube in the middle contains brown fumes. This is bromine, formed from sodium bromide. The tube on the right contains purple fumes. This is iodine, formed from sodium iodide.

With sodium chloride, the sulfuric acid behaves only as an acid, and not as an oxidising agent. This is because chloride ions have low reducing power.

With sodium bromide, the greater reducing power of bromide ions causes the sulfuric acid to be reduced, as in half-equation 1 above.

With sodium iodide, the much greater reducing power of iodide ions causes the sulfuric acid to be reduced, as in half-equations 1, 2 and 3.

CONSTRUCTING EQUATIONS

Some of the equations you see, or are asked to write, may look very complicated and be hard to remember. It is better not to try to remember them, but to work them out by the addition of half-equations. This method is recommended for the redox equations of sodium bromide and sodium iodide.

EXAM HINT

The chloride ion is behaving as a base in this reaction.

SODIUM CHLORIDE

The reaction between sodium chloride and concentrated sulfuric acid can be represented by one equation, because no redox reactions are occurring:

$$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$$

SODIUM BROMIDE

The formation of the brown fumes of hydrogen bromide can be represented by an equation analogous to the one for sodium chloride:

$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$

The table of observations shows that only one redox reaction occurs: the formation of sulfur dioxide in half-equation 1. The two relevant half-equations are:

$2Br^- \rightarrow Br_2 + 2e^-$

and

$H_2SO_4 + 2H^+ + 2e^- \rightarrow 2H_2O + SO_2$

Adding these together, then cancelling the $2e^-$ on each side, gives:

$2Br^- + H_2SO_4 + 2H^+ \rightarrow 2H_2O + SO_2 + Br_2$

You could combine the ions on the left to give:

$2HBr + H_2SO_4 \rightarrow 2H_2O + SO_2 + Br_2$

This equation represents the oxidation of the misty fumes of hydrogen bromide.

SODIUM IODIDE

The formation of the purple fumes of hydrogen iodide can be represented by an equation analogous to the one for sodium chloride:

$NaI + H_2SO_4 \rightarrow NaHSO_4 + HI$

The table of observations shows that three redox reactions occur, so the situation is more complicated. You could construct an equation showing the formation of sulfur dioxide in the same way as for sodium bromide.

Here is the result of applying the same method to the formation of sulfur in half-equation 2. The two relevant half-equations are:

$2I^- \rightarrow I_2 + 2e^-$

and

$H_2SO_4 + 6H^+ + 6e^- \rightarrow 4H_2O + S$

Before you add these together, you need to multiply the first one by three, so that the $6e^-$ on each side will cancel, giving:

$6I^- + H_2SO_4 + 6H^+ \rightarrow 4H_2O + S + 3I_2$

You could combine the ions on the left to give:

$6HI + H_2SO_4 \rightarrow 4H_2O + S + 3I_2$

This equation represents the oxidation of the misty fumes of hydrogen iodide.

You should now be able to use the same method to construct an equation to represent the oxidation of the misty fumes of hydrogen iodide to form hydrogen sulfide.

LEARNING TIP

Look carefully at all of the equations in this topic. Make sure you can identify whether they are redox reactions and, if so, what the extent of the reaction is.

CHECKPOINT

1. Explain what kind of reaction occurs when concentrated sulfuric acid is added to sodium fluoride.
2. Use half-equations to construct an overall equation for the reaction between iodide ions and concentrated sulfuric acid that results in the formation of hydrogen sulfide.

8C 4 OTHER REACTIONS OF HALIDES

SPECIFICATION REFERENCE
8.27(ii, iii)

LEARNING OBJECTIVES

- Understand precipitation reactions of the aqueous anions Cl^-, Br^- and I^- with aqueous silver nitrate solution and nitric acid, and the solubility of the precipitates in aqueous ammonia solutions.
- Understand reactions of hydrogen halides with ammonia and with water.

TESTING FOR HALIDE IONS IN SOLUTION

These tests depend on the very low solubility of silver halides in water and their different solubility in aqueous ammonia.

The reagent is silver nitrate solution, but dilute nitric acid is added first to make sure that any other anions (especially carbonate ions) are removed, as they would also form precipitates.

If a precipitate is obtained, it is then usual to add some ammonia solution. This solution can be dilute or concentrated.

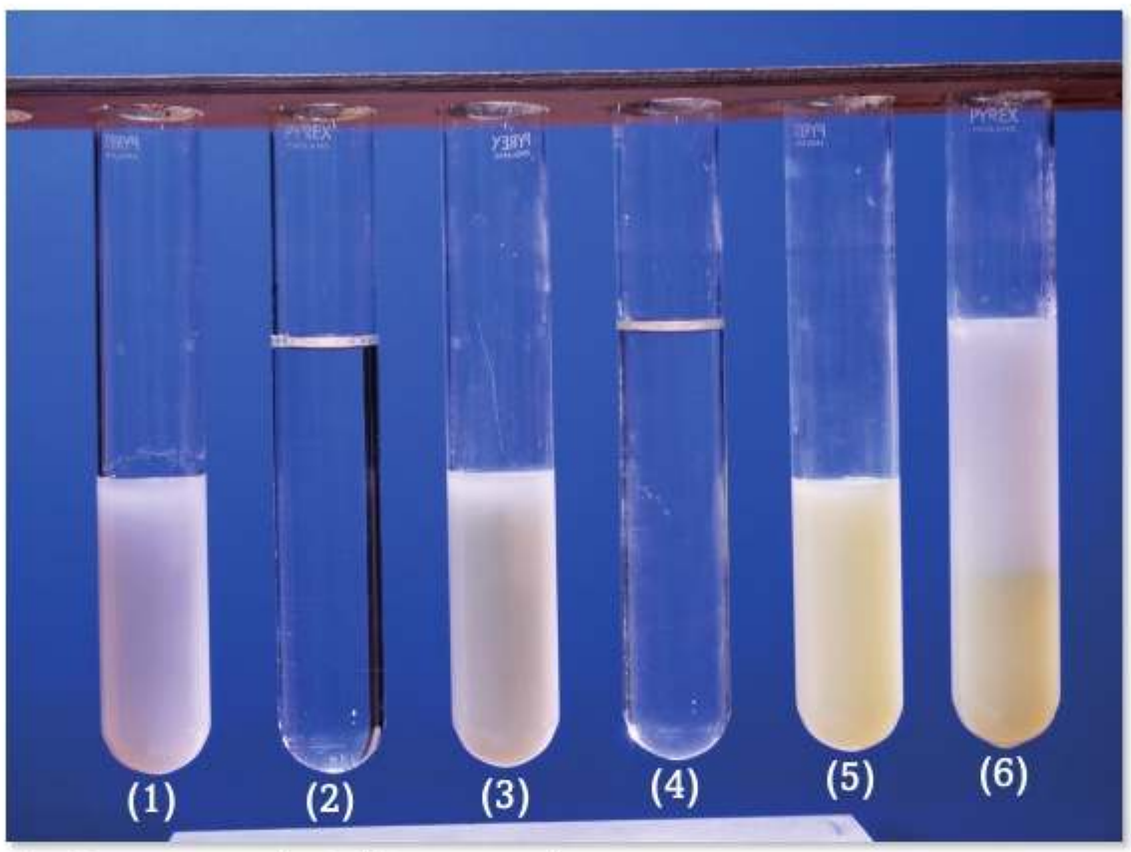
Table A and **fig A** show the results obtained. This test cannot be used to detect fluoride ions in aqueous solution, because silver fluoride is soluble.

EXAM HINT

Make sure that you can write an ionic equation for the reaction of acid with carbonate ions to produce carbon dioxide and water.

	CHLORIDE IONS	BROMIDE IONS	IODIDE IONS
add silver nitrate solution	white precipitate	cream precipitate	yellow precipitate
add dilute aqueous ammonia	soluble	insoluble	insoluble
add concentrated aqueous ammonia	soluble	soluble	insoluble

table A Results obtained from precipitation reactions of the aqueous anions Cl^-, Br^- and I^- with silver nitrate solution, followed by aqueous ammonia.



▲ **fig A** Testing for halide ions in solution.

From left to right in **fig A**:

- tube 1 shows the white precipitate formed from a chloride
- tube 2 shows the result of adding dilute aqueous ammonia to the white precipitate
- tube 3 shows the cream precipitate formed from a bromide
- tube 4 shows the result of adding concentrated aqueous ammonia to the cream precipitate
- tube 5 shows the yellow precipitate formed from an iodide
- tube 6 shows the result of adding concentrated aqueous ammonia to the yellow precipitate – it has not dissolved.

The general ionic equation for the formation of the precipitates is:

$$Ag^+(aq) + X^-(aq) \rightarrow AgX(s)$$

You can write a specific equation, such as:

$$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$$

The results in **table A** suggest that a halide ion could be identified without using aqueous ammonia. However, the colours of the three precipitates are similar. Even when all three are seen together, it is not easy to be sure which is which. In a single test, where only one precipitate is seen, this would be even more difficult.

Aqueous ammonia is a useful solvent because the precipitates have different solubilities in it.

- Silver chloride dissolves readily in both dilute and concentrated aqueous ammonia.
- Silver bromide dissolves readily in concentrated aqueous ammonia, but not in dilute aqueous ammonia.
- Silver iodide dissolves in neither dilute nor concentrated aqueous ammonia.

Dissolving of the precipitates occurs because of the formation of a complex ion. In the case of silver chloride, the equation for the reaction is:

$$AgCl(s) + 2NH_3(aq) \rightarrow [Ag(NH_3)_2]^+(aq) + Cl^-(aq)$$

HYDROGEN HALIDES ACTING AS ACIDS

All of the hydrogen halides are colourless gases and exist as polar diatomic molecules.

REACTIONS WITH WATER

The hydrogen halides readily react with water to form acidic solutions, all of which are colourless.

Table B shows these reactions.

HYDROGEN HALIDE	ACID FORMED	EQUATION
Hydrogen fluoride	hydrofluoric acid	$HF + H_2O \rightleftharpoons H_3O^+ + F^-$
Hydrogen chloride	hydrochloric acid	$HCl + H_2O \rightarrow H_3O^+ + Cl^-$
Hydrogen bromide	hydrobromic acid	$HBr + H_2O \rightarrow H_3O^+ + Br^-$
Hydrogen iodide	hydroiodic acid	$HI + H_2O \rightarrow H_3O^+ + I^-$

table B Reactions of hydrogen halides with water to form acidic solutions.

LEARNING TIP

Note the reversible arrow in the first equation. Although hydrogen fluoride is very soluble in water, it is only a weak acid, whereas the other acids formed are strong.

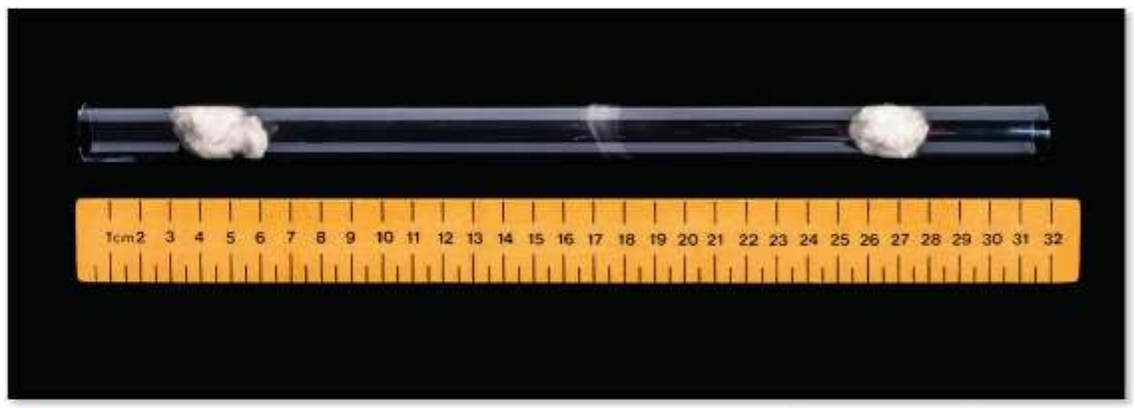
REACTIONS WITH AMMONIA

Hydrogen halides all react with ammonia gas to form salts, all of which are white ionic solids.

For example, when ammonia and hydrogen chloride gases are mixed together, they react to form ammonium chloride:

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

You may be familiar with the use of the reaction in **fig B** to illustrate diffusion and the different rates of diffusion of ammonia and hydrogen chloride.



▲ **fig B** The different rates of diffusion of ammonia (from the left) and hydrogen chloride (from the right).

Ammonia and hydrogen chloride gases are given off from the cotton wool pieces soaked in concentrated aqueous ammonia (on the left) and concentrated hydrochloric acid (on the right). These colourless gases move through the tube until they meet and react to form ammonium chloride. Ammonia gas molecules (M_r = 17.0) move more quickly and therefore travel further in the same time as hydrogen chloride molecules (M_r = 36.5). This means that the ammonium chloride does not form exactly halfway down the tube, it forms closer to the hydrochloric acid side.

FLUORIDES AND ASTATIDES

Fluorides and astatides have been mentioned in some of the reactions in this chapter, especially when it is not possible to make predictions about their reactions using trends in the chlorides, bromides and iodides. For example, you might predict that silver nitrate would form a white precipitate with sodium fluoride because sodium chloride does, but this is not the case.

If you are asked to predict reactions of fluorides and astatides that you are not familiar with, you can use information in this topic to help you. For example, you could write an equation to show the formation of an acid when hydrogen astatide is added to water, based on the equation for hydrogen iodide.

DID YOU KNOW?

None of the reactions in this topic involve redox. This is because the oxidation number of every halogen in a halogen compound in this topic remains unchanged at –1.

CHECKPOINT

1. Write equations to show:
   (a) the formation of the cream precipitate in the test for bromide ions
   (b) the precipitate dissolving in concentrated aqueous ammonia.

2. (a) Describe what happens when ammonia and hydrogen bromide gases are mixed together.
   (b) Name the product of the reaction and write an equation for its formation.